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# Cascaded sliding mode position control (SMC-PI) for an improved dynamic behavior of elastic feed drives☆

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ABSTRACT

The precision of machine tools is essentially determined by the feed drive control. To improve the dynamic behavior of elastic feed drives, this paper presents a novel position control concept, substituting the conventional P position controller of the industrial cascade control with a continuous sliding mode controller (SMC) together with a Kalman Filter for state estimation. The novelty of this approach lies in the focus on practical applicability: The underlying PI-cascade structure leads to low commissioning effort and the SMC controller's parameterization is simplified to a single tuning parameter. A general compliant feed drive model is used as plant, whose parameters usually are already known from commissioning of the velocity controller. The SMC position controller is derived analytically and its stability and damping properties are investigated for the dominating mechanical mode for uncertainties of the model parameters and a wide range of controller gains. The effectiveness of the SMC-PI controller is validated for an industrial ball screw feed drive. Extensive experimental results show a considerable decrease of tracking errors in command tracking, disturbance rejection and for a high speed cutting (HSC) milling experiment compared to P-PI cascade control, while stability margins remain almost independent of the controller gains. The parameterization is discussed on the basis of a theoretical analysis and measuring results. Furthermore, excellent robustness properties regarding model errors are demonstrated.

## 1. Introduction

In the field of machine tools and manufacturing units, feed drive systems are a key component for the satisfactory execution of manufacturing tasks. The nominal paths provided by the Numerical Control (NC) are converted by the individual machine axes into a relative movement between the tool and workpiece. In addition to high accuracy requirements to ensure the desired workpiece quality, increasingly high dynamic capabilities are expected of the feed drives. For instance, by implementing high feed velocities and accelerations, the machining time can be reduced and productivity consequently increased [1]. The motion generation of modern manufacturing units is nowadays mainly achieved through the use of electromechanical feed drives with a ball screw as a mechanical transmission element. Such drive systems are operated almost exclusively with the aid of cascaded control loops, whereby an integral proportional controller (PI) is used for the speed control loop and a proportional controller (P) for the position control loop. There are several reasons for the dominant position of these so-called P-PI cascade controllers. On the one hand, the parameters of the individual control loops can simply be parameterized from the inner control loop to the outer control loop, which simplifies the parameterization by only having to tune one simple controller at a time. On the other hand, such control systems are robust against parameter changes of the system and can be implemented cost-effectively. However, the maximum achievable gain in the position control loop ($K_v$ factor) and, thus, also the bandwidth is strictly limited due to the structural elasticities of the coupled mechanical elements.

Within this paper a SMC position controller is presented, which is capable of dealing with the compliant mechanics of feed drives and cannot be addressed by a P controller. The usually underlying speed and current control cascades are kept. The novel contribution is the aim to narrow the gap between academic controllers, where the tuning complexity in the commissioning phase typically outweighs the performance benefits, and the current state of the industry, P-PI cascaded control. Therefore, the SMC is consequently brought down to one gain parameter, comparable to P position control and the controller is implemented and shown to perform well without additional compensation measures, which needed to be tuned as well. The underlying

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drive model is chosen as a generic $PT_2$ model, which can be applied to general compliant feed drives and is not just limited to ball-screw drives.

The paper is structured as follows: The state of the art of electromechanical feed drive control is introduced in Section 2, where we conclude, why extending the position controller to SMC while retaining the cascaded structure is a logical consequence of elastic feed drives. In Section 3 the controller design is presented for a controllable single input single output (SISO) system, which is applied in Section 4 to an elastic feed drive model by approximating the first mode. The parameterization of the position controller is theoretically analyzed in Section 5, taking model uncertainties into account. Furthermore, the control gains are reduced to one tuning parameter, which makes the parameterization effort comparable to the one of a P controller. Finally, in Section 6 results are presented for the novel controller in comparison with state-of-the-industry P-PI cascaded control in time and frequency domain for command tracking as well as disturbance rejection.

## 2. State of the art

Approaches for improving the dynamics of feed drives can be classified in three categories: Advancements in the mechanical structure, in the input trajectory and in the drive control. Whereas the research described within the following sections takes place in the latter field of drive control, the three categories usually complement each other.

Mechanical approaches pursue the contradictory goal to increase the stiffness under static and dynamic conditions while reducing the masses and inertias being moved. This is achieved, e.g., by active variation of the preloading [2], by damping the axial bearings [3], or by spindle vibration compensation with tuned mass dampers in the spindle shaft [4]. Advancements in providing better input trajectories aim to avoid excitation of unwanted dynamics by using higher-order differentiable (i.e. smoother) input signals [5] or by filtering them [6], to directly incorporate a dynamics model as constraints in path planning [7] or to additively precompensate predicted errors [8]. Although changes in the input trajectory positively influence the trajectory following behavior, they cannot change the static and dynamic drive stiffness and, hence, have no influence on disturbance suppression.

Various higher order controllers were investigated as replacements for cascaded control of feed drives, but never found their way into industrial application due to their increased complexity in commissioning. Erkorkmaz and Altintas propose a state controller in combination with a Kalman filter for disturbance observation and a zero phase error tracking controller (ZPETC) [9]. This controller is extended by Gordon and Erkorkmaz with explicit consideration of a flexible drive model, which is shown, in combination with a prefiltered trajectory, to reduce dynamic tracking errors in comparison to P-PI cascaded control [10]. The higher modeling depth and the trajectory prefilter however require precise tuning of 17 parameters. Sepasi et al. explicitly consider varying load masses and table positions in a linear parameter varying (LPV) model, which they use for a gain scheduling controller approach [11]. For tuning of the LPV model this approach however needs an identification of the varying parameters, which is performed in 5 mm steps along the spindle and leads to an increased commissioning effort. Uchiyama shows an adaptive control approach with feedforward and feedback filters [12]. Zhang and Chen use an active disturbance rejection controller (ADRC) for motor control and model the elasticities between motor and table with an equivalent error model for ball screw, which they show to perform better than P-PI cascaded control [13]. Papageorgiou et al. apply an adaptive backstepping controller (ABSC) with the scope of resiliency against varying friction parameters [14]. They also show their controller to outperform the P-PI cascaded control. Both, ADRC and ABSC show good trajectory tracking and disturbance rejection, but for the cost of increased commissioning effort, e.g., 10 parameters for ADRC [13], or 19 parameters for the ABSC [14], compared to cascaded control with three parameters for the position and speed cascades. While adaptive approaches, as the ABSC promise to reduce the commissioning effort by automatically tuning the required parameters, they generally move the problem to finding suitable objective weights, which essentially determine the reachable performance of the controllers and need to be tuned as well.

With sliding mode control one controller type received special attention due to its inherent robustness properties and its guaranteed finite-time convergence. However, the variable system structure in pure SMC together with finite sample times results in chattering of the output, which is caused by the switching function in the control law. Typical countermeasures to eliminate this, usually unwanted, chattering are smoother switching functions (equivalent control), using a second-order sliding surface (super-twisting SMC), or lowpass-filtering the control signal [15–17]. Altintas et al. introduce an SMC with adaptive disturbance observer based on a rigid drive model, which shows comparative tracking behavior than state control with pole placement, but better disturbance cancellation [18]. Their approach neglects the compliant drive behavior and, hence is only validated for slow motions. Erkorkmaz and Kamalzadeh extend this controller with friction compensation and notch filters to dampen the rotational modes of the ball screw spindle [19], further with Kalman-estimated harmonic motor ripple compensation [20] and applied it to linear direct drives by Altintas and Okwudire [21]. Yet, the additional compensation and filter structures need to be parameterized carefully to not deteriorate the system's performance. As the vibrations caused by the drive's compliance are a limiting factor for increased controller dynamics, Kamalzadeh and Erkorkmaz explicitly model the first mode of axial vibration [22] and Okwudire and Altintas consider a flexible 2-degrees of freedom (DOF) model to design a mode-compensating single input multi output controller, which dampens the first mode of the feed drive during tracking control [23]. Both approaches add further model complexity and additionally need tuning of compensation filters. By using a nonlinear sliding surface Mohammad et al. are able to reduce the controller's energy consumption without degrading its tracking performance [24]. Yau and Yan show, that an equivalent-control SMC with a bristle friction model can also be used for nano-precision control of ball screw drives [25], which adds 13 parameters for the friction model alone. An adaptive SMC approach for nonlinear, underactuated systems is presented by Muske et al., where a model predictive controller (MPC) is used to adapt the sliding surface of a direct SMC [26]. The controller performance turns out similar to offline optimization of the sliding surface parameters for an inverted pendulum. The stability under disturbances, however, is only investigated for the SMC, but together with the MPC, the stability for both controllers together is only shown for the nominal case.

In contrast to higher order direct controllers, also extensions and improvements to cascaded control were investigated. Pritschow et al. show, that the dynamic stiffness of cascaded control can be increased by feeding back the machine table acceleration measured with Ferraris sensor in the current control cascade [27]. Further, Verl and Frey present a semiactive damping approach by adding a frictional actuator to table slide, which allows to increase the dynamics due to better damped mechanics [28]. Both approaches, however, require additional hardware for sensing or actuation and come with additional costs. Sun et al. show significant improvements of >80 % in bandwidth for cascaded control by introducing changes in the control structure, such as adding a second P velocity control loop (one for motor speed and one for table velocity) [29] and by realizing the position controller as PD-controller [30]. However, both approaches are constrained to the special case of systems with large inertia ratio between the table mass and the equivalent rotatory masses, e.g, due to large spindle pitch or very high load masses. Papageorgiou et al. replace the PI speed controller with three different higher order controllers, i.e. two adaptive controllers and one super-twisting SMC, in [31]. For an experimental

setup consisting of drive and load permanent-magnet synchronous motors (PMSMs) with adjustable friction they show that the mean absolute error can be especially well reduced by the SMC in combination with an adaptive friction observer compared to a PI speed controller. For the SMC speed controller, they require an identified two-mass model and a tuned linear matrix inequality (LMI) with properly chosen tuning gains. The adaptive approaches move the parameter identification problem to a problem of tuning the objective function's weights. Pisano et al. also enhance the speed cascade and, additionally, the current controller by using a second-order SMC for each cascade, which reduces the commissioning effort to one parameter for each controller [32]. The approach is investigated for speed control only and, hence, the behavior of such a control setup for compliant drive axes being coupled with the motor is not addressed. Guo et al. propose a cascaded SMC position controller for a hydraulic parallel robot [33]. Their approach shows slight improvements for tracking control compared to a P controller. The used approach, however, is tailored for the nonlinear pressure dynamics of the hydraulic actuators and there is no underlying speed controller, such that this approach cannot be directly compared with cascaded drive control. Finally, Tran et al. show an approach of cascaded SMC-PID control [34]. The authors show the merits of this approach and use a $PT_2$ model as approximation of the closed loop velocity plant. Their goal is to reduce the step response overshoot in contrast to trajectory following control and the approach is exclusively evaluated in simulation. Finally, Petronijević et al. use a cascaded control approach, where every controller is replaced by a discrete-time SMC and a nonlinear disturbance compensation is added [35]. The authors present good tracking and disturbance behavior and prove excellent robustness for matched uncertainties, yet, this approach leaves roughly 10 parameters to tune per cascade.

The control manufacturer Fanuc also offers a control extension for machines with a large mass inertia ratio with a additional functionality called Dual Position Feedback [36]. A learning control and a filter circuit reduce the positioning error by considering the motor as well as the table position. Siemens distributes the method Advanced Position Control (APC), in which the damping of the natural frequencies is increased by an additional feedback of the table speed [37]. The control signal in the speed control loop is synthesized from the motor and the table speed via a parallel connection of several band-pass filters. In this way, the amplitude reserve in the position control loop can be significantly increased, which allows the $K_v$ factor to be amplified. The method, however is needs to exactly match the eigenfrequencies and is not robust against changes in the plant, e.g. due to wear or temperature. The controller specialist Heidenhain [38] meets the increasing demands in milling with the additional Computerized Numerical Control (CNC) functions PAC (Position Adaptive Control) and MAC (Motion Adaptive Control). Here, a position or speed-dependent adaptation of the cascade control takes place, which increases the path accuracy. This gain adaption is applicable to other control methods as well.

While mostly described as part of control approaches, several methods for compensation of non-ideal effects, such as runout [11,39], backlash [40,41], disturbance forces [18,42] and friction [43,44], were investigated instead of directly including them in the controller. These methods can mostly be applied independently of the used controller, although, some controller approaches are reported to be sensitive against missing compensation measures (e.g., [22,23]).

On top of the search for better controllers and control structures the controller parameterization has become a partly separate research topic. While for cascaded control multiple heuristics [45] and model-based tuning rules [46,47] exist, which already lead to good performance, the problem becomes harder for more involved controller structures. But also for cascaded control the parameters from these rules are usually rather conservative if the plant has varying parameters or dominating nonlinearities. Hence, optimal parameterization is examined in various works for cascaded control as $H_\infty$ problem [48,49] and also applied to other control structures [50].

The state of the art shows that direct, higher order controllers are able to improve the dynamic behavior of feed drives, especially for dominating elasticities. However, they did not find their way into industrial hardware, especially due to their increased modeling depth and, hence, their increased commissioning effort. Adaptive controllers evade this problem by automatic adaption of some or all parameters, but introduce new parameters that need to be tuned for proper convergence rates. Besides, with the online adaption of parameters the controller may behave non-predictable during runtime by changing its behavior and yielding different results for the same trajectory. For cascaded controllers the state of the art reveals two general approaches: extending the cascaded structure, or applying more advanced controllers to one or all of the controller cascades. The latter approach mostly shows better performance, but also requires a deeper understanding of the plant model and, hence, is more difficult to parameterize. We argue, that the remaining degrees of freedom of the compliant feed drive mechanics correspond with the position controller, as the speed control loop is closed with the motor encoder. Hence, we focus on enhancing only the position controller with a SMC, while retaining the cascaded structure along with its established tuning rules and robustness properties. The parameterization is brought down to one controller gain and two plant parameters, which only have to be estimated approximately due to the robustness of the controller. This way, the gap between high sophisticated but complex control structures and the simple but less performant P-PI approach shall be closed.

## 3. Sliding mode controller design

The characteristic feature of all SMC methods is based on the achievement of special sliding modes, in which the dynamics of the closed loop is determined by the selected sliding surface. Since the sliding modes are largely independent of parameter changes of the plant and external disturbances, robust controllers with excellent dynamic properties can be obtained. The starting point for the design of the sliding mode controller is a linear time-invariant SISO system of $n$th order, whose dynamics can be described using the state space representation

$$\tilde{\Sigma} : \begin{cases} \dot{\tilde{\boldsymbol{x}}}(t) = \tilde{\boldsymbol{A}}\tilde{\boldsymbol{x}}(t) + \tilde{\boldsymbol{b}}u(t) \\ y(t) = \tilde{\boldsymbol{c}}^{\mathrm{T}}\tilde{\boldsymbol{x}}(t) \end{cases} , \tag{1}$$

where $\tilde{\boldsymbol{x}}(t)$ denotes the state vector, $\tilde{\boldsymbol{A}}$ the system matrix, $\tilde{\boldsymbol{b}}$ the input vector and $u(t)$ the control signal. As a prerequisite, it must be ensured that the considered system model $\tilde{\Sigma}$ is available in the so-called controllable canonical form. If this is not the case, a linear state transformation

$$\boldsymbol{x}(t) = \boldsymbol{T}^{-1}\tilde{\boldsymbol{x}}(t) \tag{2}$$

of the real physical system into controllable canonical coordinates $\tilde{\Sigma} \rightarrow \Sigma$ must take place at first as described in [51,52]. This transformation is only possible if the system is controllable, i.e. the associated controllability matrix

$$\boldsymbol{S}_0 = \left[\tilde{\boldsymbol{b}},\ \tilde{\boldsymbol{A}}\tilde{\boldsymbol{b}},\ \tilde{\boldsymbol{A}}^2\tilde{\boldsymbol{b}}, \dots,\ \tilde{\boldsymbol{A}}^{n-1}\tilde{\boldsymbol{b}}\right] \tag{3}$$

has full rank. If the last row of the inverse controllability matrix is designated with

$$\boldsymbol{s}^{\mathrm{T}} = [0\ 0 \dots 0\ 1]\,\boldsymbol{S}_0^{-1}, \tag{4}$$

the necessary transformation matrix can be set up using the relationship

$$\boldsymbol{T} = \begin{bmatrix} \boldsymbol{s}^{\mathrm{T}} \\ \boldsymbol{s}^{\mathrm{T}}\tilde{\boldsymbol{A}} \\ \boldsymbol{s}^{\mathrm{T}}\tilde{\boldsymbol{A}}^2 \\ \vdots \\ \boldsymbol{s}^{\mathrm{T}}\tilde{\boldsymbol{A}}^{n-1} \end{bmatrix}^{-1} . \tag{5}$$

Subsequently, the system can be specified in controllable canonical form

$$\Sigma : \begin{cases} \dot{\boldsymbol{x}}(t) = \boldsymbol{T}^{-1}\tilde{\boldsymbol{A}}\boldsymbol{T}\boldsymbol{x}(t) + \boldsymbol{T}^{-1}\tilde{\boldsymbol{b}}u(t) \\ y(t) = \tilde{\boldsymbol{c}}^{\mathrm{T}}\boldsymbol{T}\boldsymbol{x}(t) \end{cases} . \tag{6}$$

For a compact notation, the following relationships are introduced:

$$\begin{aligned} \boldsymbol{A} &= \boldsymbol{T}^{-1}\tilde{\boldsymbol{A}}\boldsymbol{T}, \\ \boldsymbol{b} &= \boldsymbol{T}^{-1}\tilde{\boldsymbol{b}}, \\ \boldsymbol{c}^{\mathrm{T}} &= \tilde{\boldsymbol{c}}^{\mathrm{T}}\boldsymbol{T}. \end{aligned} \tag{7}$$

This is also the starting point, if the considered system already exists in controllable canonical form.

Next, a sliding surface in the $n$-dimensional error space needs to be defined, whereby the commonly used approach of the form

$$S\left(e_x, \dot{e}_x, \ldots, \overset{(n-1)}{e}_x\right) = \left(\frac{\mathrm{d}}{\mathrm{d}t} + \lambda\right)^{n-1} e_x \tag{8}$$

is used. Therein, $\lambda$ denotes a setting parameter and the error term

$$e_x = x_{\mathrm{s}} - x_{\mathrm{a}} \tag{9}$$

measures the control deviation between the setpoint $x_{\mathrm{s}}$ and the actual value $x_{\mathrm{a}}$. For a clearer representation, the sliding surface can also be written in vector representation

$$S(\boldsymbol{e}_x) = \boldsymbol{\Lambda}(\lambda)^{\mathrm{T}} \begin{bmatrix} x_{\mathrm{s}} - x_{\mathrm{a}} \\ \dot{x}_{\mathrm{s}} - \dot{x}_{\mathrm{a}} \\ \vdots \\ \overset{(n-1)}{x_s} - \overset{(n-1)}{x_a} \end{bmatrix} = \boldsymbol{\Lambda}(\lambda)^{\mathrm{T}} \underbrace{\left(\boldsymbol{x}_{\mathrm{s}} - \boldsymbol{x}_{\mathrm{a}}\right)}_{\boldsymbol{e}_x}, \tag{10}$$

where $\boldsymbol{\Lambda}(\lambda)$ are the coefficients of the characteristic polynomial, which also result from Eq. (8).

The basic principle for designing an asymptotic tracking controller can now be divided into two steps. Initially, a control signal $u_{SM}$ is determined, which ensures that the trajectory remains on the sliding surface until the steady-state is reached. For this purpose, the necessary and sufficient condition

$$\dot{S} = \boldsymbol{\Lambda}(\lambda)^{\mathrm{T}}\dot{\boldsymbol{e}}_x = \boldsymbol{\Lambda}(\lambda)^{\mathrm{T}}\left(\dot{\boldsymbol{x}}_{\mathrm{s}} - \dot{\boldsymbol{x}}_{\mathrm{a}}\right) \overset{!}{=} 0 \tag{11}$$

must be satisfied. Inserting the state space model (6), (7) leads to

$$\boldsymbol{\Lambda}(\lambda)^{\mathrm{T}}\left(\dot{\boldsymbol{x}}_{\mathrm{s}} - \left(\boldsymbol{A}\boldsymbol{x}_{\mathrm{a}} + \boldsymbol{b}u\right)\right) = 0, \tag{12}$$

from which the control variable in the sliding mode phase

$$u_{SM} = \left(\boldsymbol{\Lambda}(\lambda)^{\mathrm{T}}\boldsymbol{b}\right)^{-1}\boldsymbol{\Lambda}(\lambda)^{\mathrm{T}}\left(\dot{\boldsymbol{x}}_{\mathrm{s}} - \boldsymbol{A}\boldsymbol{x}_{\mathrm{a}}\right) \tag{13}$$

results. Subsequently, the obtained control law is extended by a component $u_{\mathrm{RC}}$, which ensures that the initial state is transferred to the sliding surface ($S = 0$). In order to satisfy this so-called reachability condition, the energy of the system must decrease over time, according to the stability theory of Lyapunov. By choosing the Lyapunov function candidate

$$V(\boldsymbol{e}_x) = \frac{1}{2}S^2 = \frac{1}{2}\boldsymbol{e}_x^{\mathrm{T}}\boldsymbol{\Lambda}(\lambda)\boldsymbol{\Lambda}(\lambda)^{\mathrm{T}}\boldsymbol{e}_x > 0 \quad \forall \boldsymbol{e}_x \in \mathbb{R}^n \setminus \{0\} \tag{14}$$

it can be seen that the systems energy decreases if the condition

$$\dot{V}\left(\boldsymbol{e}_x, \dot{\boldsymbol{e}}_x\right) = S\dot{S} = \boldsymbol{e}_x^{\mathrm{T}}\boldsymbol{\Lambda}(\lambda)\boldsymbol{\Lambda}(\lambda)^{\mathrm{T}}\dot{\boldsymbol{e}}_x < 0 \tag{15}$$

is fulfilled. To achieve this condition, discontinuous switching functions (e.g. the sign-function) are often used in SMC. However, the resulting switching noise (chattering) leads to high-frequency oscillations around the equilibrium state, so that fast dynamic components can be excited in real systems [53]. In order to prevent this effect, which is typical for SMC, the approach

$$\dot{V} = \frac{\mathrm{d}V}{\mathrm{d}t} = S\dot{S} \overset{!}{=} -KS^2 < 0 \quad \forall \boldsymbol{e}_x \in \mathbb{R}^n \setminus \{0\} \tag{16}$$

with the positive constant $K$ is chosen, which guarantees globally exponential stability. The resulting relationship

$$\dot{S} = -KS \tag{17}$$

can then be transformed with (6), (7) and (8) to

$$\boldsymbol{\Lambda}(\lambda)^{\mathrm{T}}\left(\dot{\boldsymbol{x}}_{\mathrm{s}} - \left(A\boldsymbol{x}_{\mathrm{a}} + \boldsymbol{b}u\right)\right) = -KS. \tag{18}$$

By solving this for the plant input, the complete control law

$$\begin{aligned} u &= u_{\mathrm{SM}} + u_{\mathrm{RC}} \\ &= \left(\boldsymbol{\Lambda}(\lambda)^{\mathrm{T}}\boldsymbol{b}\right)^{-1}\boldsymbol{\Lambda}(\lambda)^{\mathrm{T}}\left(\dot{\boldsymbol{x}}_{\mathrm{s}} - A\boldsymbol{x}_{\mathrm{a}} + K\left(\boldsymbol{x}_{\mathrm{s}} - \boldsymbol{x}_{\mathrm{a}}\right)\right) \end{aligned} \tag{19}$$

is obtained, where now the necessary term to ensure the reachability condition

$$u_{\mathrm{RC}} = \left(\boldsymbol{\Lambda}(\lambda)^{\mathrm{T}}\boldsymbol{b}\right)^{-1}\boldsymbol{\Lambda}(\lambda)^{\mathrm{T}}K\left(\boldsymbol{x}_{\mathrm{s}} - \boldsymbol{x}_{\mathrm{a}}\right) \tag{20}$$

is included. Note that the derived control law (19) can be applied to any linear controllable system model, giving the user two degrees of freedom ($K$, $\lambda$) for control optimization.

## 4. SMC-PI cascade control for elastic feed drive systems

The control law (19) is used to replace the traditional P controller. Since the cascade structure is not being modified, the inner control cascades can be parameterized as usual without any changes. To simplify the parameterization effort the plant model for the SMC position controller, consisting of the closed-loop velocity plant and the compliant drive mechanics, is chosen as simple as possible, while containing the dominant compliance. This results in larger model errors, but also less parameterization effort and a smaller number of plant parameters. Sliding mode controllers are known for their robustness against such model errors. Thus, it is sufficient to abstract the drive system (ball-screw drive, rack and pinion drive, etc.) as a single-mass oscillator. For this purpose, an oscillatory second order lag element ($\mathrm{PT}_2$) with a subsequent integrator (I) is used, resulting in the 3rd order dynamic system ($\mathrm{PT}_2\mathrm{I}$)

$$\begin{bmatrix} \dot{x}(t) \\ \ddot{x}(t) \\ \dddot{x}(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -\omega_0^2 & -2D\omega_0 \end{bmatrix} \begin{bmatrix} x(t) \\ \dot{x}(t) \\ \ddot{x}(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \omega_0^2 \end{bmatrix} u(t). \tag{21}$$

The model only requires two parameters, damping $D$ and the characteristic angular frequency $\omega_0$, which can be determined from measurements on the feed drive (e.g. frequency response) and are usually already known from systematic commissioning of the speed control loop. At the same time, this approach neglects higher order plant dynamics as well as ball-screw specific effects, such as the varying spindle stiffness, which act as disturbances on the position controller. The application of the control law according to (19) to the controlled plant (21) finally yields the relationship

$$\begin{aligned} u = \frac{1}{\omega_0^2}\Big(&2D\omega_0\ddot{x}_{\mathrm{a}} + \omega_0^2\dot{x}_{\mathrm{a}} + \dddot{x}_{\mathrm{s}} + (2\lambda + K)\,\ddot{e}_x \\ &+ \left(\lambda^2 + 2\lambda K\right)\dot{e}_x + \left(\lambda^2 K\right)e_x\Big). \end{aligned} \tag{22}$$

Note, it is possible to add more terms to this law, e.g. for disturbance estimation. This could lead to further performance advantages. However, the idea of this paper is to keep the terms as simple as to limit the parameterization effort. The structure of the entire system is shown in Fig. 1. The application of the positioning control law (22) requires the planning of sufficiently smooth reference trajectories $x_{\mathrm{s}}(t) \in C^3$. Usually, an s-shaped profile is used to smooth the path planning in the area of NC machine tools, where the jerk passes through 7-phases with limited nominal values, hence, the condition is fulfilled a priori. In contrast to the P-position control loop, the actual velocity as well as actual acceleration of the machine table must be fed back in addition to the table position. For the table velocity, the rotational speed of the motor is taken from the speed control loop and converted into a translational velocity using the pitch of the ball screw. A particular challenge is the acquisition of the actual acceleration signal. The easiest way is to attach an additional acceleration sensor to the machine table, whereby the integration of additional sensors is associated with effort and costs. To avoid this, the acceleration signal can be reconstructed by any state observer, e.g., a Kalman filter [54] or a sliding mode observer (SMO) [55].

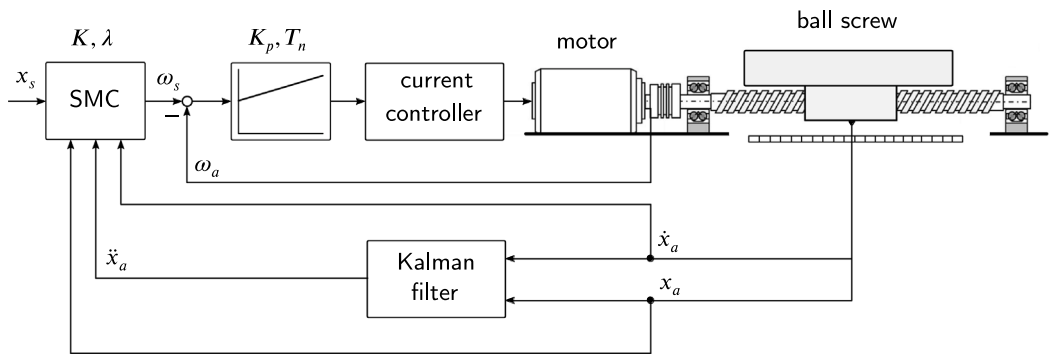



**Fig. 1.** Structure of the SMC-PI position control system.

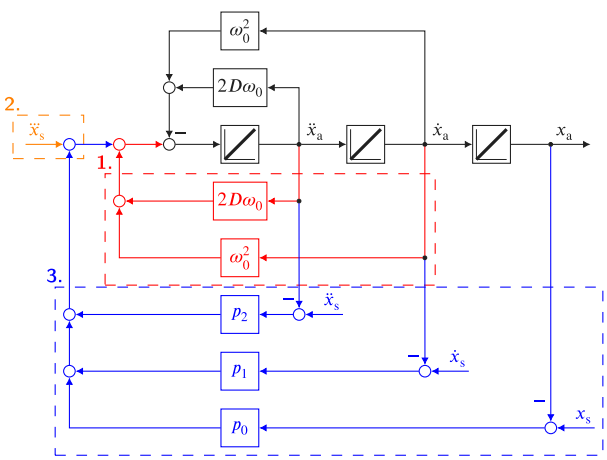



**Fig. 2.** Signal flow chart of the SMC-PI cascade control.

## 5. Controller parameterization

Before the parameterization is discussed, it is useful to examine the qualitative influence of $K$ and $\lambda$. The control law (22) can be subdivided into three parts, see Fig. 2.

The *compensation* term

$$u_1 = \frac{1}{\omega_0^2}\left(2D\omega_0\ddot{x}_a + \omega_0^2\dot{x}_a\right) \tag{23}$$

incorporates the current table speed and acceleration into the control loop. Due to the weighting of the states with the values of the $PT_2$ element (characteristic angular frequency $\omega_0$ and damping $D$) the compensation term corresponds to the exact inversion of the system's transfer behavior. Consequently, a chain of $n = 3$ integrators remains as the controlled system.

By introducing the *feedforward* term

$$u_2 = \frac{1}{\omega_0^2}\dddot{x}_s, \tag{24}$$

i.e. the jerk setpoint value, an ideal transfer behavior would result from the threefold time integration, assuming an ideal system model and the absence of disturbances. Since these idealized conditions do not exist in practice, the two terms are extended by the real *feedback* term

$$u_3 = \frac{1}{\omega_0^2}\left(p_2\ddot{e}_x + p_1\dot{e}_x + p_0 e_x\right). \tag{25}$$

In this way, the deviation of the respective states between actual value and setpoint is weighted with the corresponding coefficients

$$\begin{aligned} p_2 &= 2\lambda + K, \\ p_1 &= \lambda^2 + 2\lambda K, \\ p_0 &= \lambda^2 K. \end{aligned} \tag{26}$$

Substituting the control law (22) in the system formulation (21) yields the error dynamics

$$\dddot{e}_x + p_2\ddot{e}_x + p_1\dot{e}_x + p_0 e_x = 0, \tag{27}$$

which determines the transient behavior during positioning with tracking error ($e_x = x_s - x_a$). By setting up the characteristic equation, the poles of the error dynamics can be calculated as $s_{1,2} = -\lambda$ and $s_3 = -K$. The double pole is assigned to the oscillatory $PT_2$-element. Since it has no imaginary component, the error dynamics and thus the transient response is always aperiodically damped, regardless of the selection of the setting parameters. Consequently, the SMC inherently meets the requirement for positioning without overshoot in the position control loop. The error dynamics and thus the position control loop is always stable if control parameters $K > 0$ and $\lambda > 0$ are selected. In order to achieve a fast control loop, $K$ and $\lambda$ should be chosen sufficiently large to force a fast decay of the tracking error. In practical terms, however, the parameters of the controller are limited because poles far in the left half plane will lead to almost differentiating behavior, which amplifies noise and disturbances. As the smallest poles essentially characterize the system dynamics it is further chosen that $K \equiv \lambda$, which also reduces the number of degrees of freedom in the controller parameters to one controller gain. This facilitates the parameterization for commissioning engineers to a level that is comparable to $K_v$ gain tuning, although it is a higher order controller. For a concise notation, the controller gain is written as $\lambda$ from now on, where also $K$ is affected.

### 5.1. Uncertainty analysis

To determine the robustness of the proposed controller against errors in the plant parameters $D$ and $\omega_0$, e.g. resulting from the position-dependent stiffness of the ball screw spindle or variations of the load mass, we consider the non-ideal controller

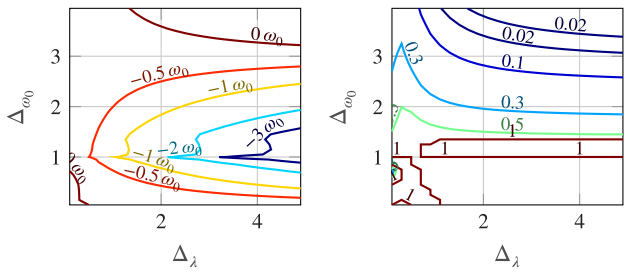
$$\begin{aligned} \hat{u} = \frac{1}{\hat{\omega}_0^2}\Big(&2\hat{D}\hat{\omega}_0\ddot{x}_a + \hat{\omega}_0^2\dot{x}_a + \\ &\dddot{x}_s + p_2\ddot{e}_x + p_1\dot{e}_x + p_0 e_x\Big) \end{aligned} \tag{28}$$

with $\hat{D} = \Delta_D D$ and $\hat{\omega}_0 = \Delta_{\omega_0}\omega_0$ being the disturbed plant parameters with multiplicative disturbances $\Delta_D$ and $\Delta_{\omega_0}$. Using this controller with the system (21) leads to the disturbed system dynamics

$$\begin{aligned} \dddot{x}_a = \frac{1}{\Delta_{\omega_0}}\Big(&2D\omega_0(\Delta_D\Delta_{\omega_0} - \Delta_{\omega_0}^2)\ddot{x}_a + \\ &\dddot{x}_s + p_2\ddot{e}_x + p_1\dot{e}_x + p_0 e_x\Big). \end{aligned} \tag{29}$$

For a stability analysis the disturbed system dynamics can be transformed into a frequency transfer function between desired and actual position

$$\begin{aligned} G_\Delta(s) = \frac{s^3 + p_2 s^2 + p_1 s + p_0}{\Delta_{\omega_0}^2 s^3 + \Big(2D\omega_0(\Delta_D\Delta_{\omega_0} - \Delta_{\omega_0}^2) + p_2\Big)s^2 + p_1 s + p_0}. \end{aligned} \tag{30}$$




(a) Stability via real part of the dominant pole in multiples of $\omega_0$

(b) Controller damping ratio $\xi$ of the dominant pole (pair)

**Fig. 3.** Stability and damping analysis of the proposed control law with multiplicative variation in the controller gain and the characteristic angular frequency.

**Fig. 4.** Test bench with ball screw feed drive.

Note, that this transfer function reduces to a unit transfer function and, hence, its poles cancel out for $\Delta_D = \Delta_{\omega_0} = 1$. Further, the damping disturbance $\Delta_D$ has only minor influence on the poles of $G_\Delta$ via the damping compensation term $2D\omega_0(\Delta_D\Delta_{\omega_0} - \Delta_{\omega_0}^2)$ and can be neglected. The stability of $G_\Delta$ is analyzed via the location of its dominant pole/pole pair depending on $\Delta_{\omega_0}$ and the controller gain $\lambda$. For the stability analysis $\lambda$ is chosen relative to $\omega_0$, i.e., with a factor $\Delta_\lambda := \lambda/\omega_0$. The poles can only effectively analyzed numerically. Without loss of generality the system parameters are chosen as $D = 0.5$ and $\omega_0 = 1$ rad/s for the remainder of the robustness analysis, the qualitative behavior does not change for different choices of parameters, as the variation of poles $\Delta_\lambda$ and characteristic frequency $\Delta_{\omega_0}$ both relatively depend on $\omega_0$. Fig. 3a shows the real part of the dominant eigenvalue. As expected, with increasing controller gain (increasing $\Delta_\lambda$), the dominant eigenvalue becomes faster. This is especially the case for $\omega_0$ being perfectly known ($\Delta_{\omega_0} = 1$). Further, for the displayed variation of controller parameters, the controller remains stable for an overestimation of $\hat{\omega}_0 \leq 3\omega_0$, except for the lower left corner of Fig. 3a, where $\omega_0$ is underestimated and, additionally, the controller gain is small. Fig. 3b shows the damping factor $\xi := |\Re(s_D)|/\|s_D\|$ of the dominating pole $s_D \in \mathbb{C}$, which effectively describes the overshoot of the trajectory following error for varying $\Delta_{\omega_0}$. It reveals the overshooting-free positioning behavior $\xi = 1$ for $\Delta_{\omega_0} = 1$, which results from the error differential equation (27). Further, the damping stays good with $\xi \approx 1$ for $\Delta_{\omega_0} \in [0, 1.35)$ and rapidly worsens for a bigger overestimation of $\omega_0$. This becomes even steeper for an increasing controller gain.

In total, from a robustness perspective it is advisable to choose the controller gain $\lambda > 0.5\omega_0$, but in the range of good model certainty since the $PT_2I$ approximation typically becomes more uncertain for higher frequencies. For convergence times it is better to slightly overestimate $\omega_0$, as the gradients in Fig. 3a rise slower for $\Delta_{\omega_0} > 1$ than for $\Delta_{\omega_0} < 1$, whereas for good damping behavior a smaller value of $\hat{\omega}_0$ is preferable.

## 6. Experimental results

To verify the derived SMC-PI controller, it is implemented and investigated on a ball screw feed drive of a machine tool (see Fig. 4). For easier accessibility and control openness, this part was extracted from the machine tool and set up as a test bench on a machine bed. However, the components used and their dynamic behavior are equivalent to the industrial setup of a machine tool axis. Table 1 summarizes the relevant parameters of the test bench. The current control loop as well as current setpoint filters are running on the Siemens inverter. The velocity and position control loops as well as the setpoint generation were outsourced to a dSPACE rapid prototyping system with a cycle time of 4 kHz. The structure and functionality of these functions are equivalent to the industrial standard. The separation allows free configurability, so that other controllers than the P-PI cascade control can be implemented. This setup has only the disadvantage of an additional communication dead time of ~1 ms for reading the current controller setpoints and writing them on the internal field bus. This affects both the presented SMC-PI controller and the reference controller (P-PI), so the comparison is still fair.

The parameterization of the underlying current control loop is carried out by means of the auto-tuning function of the Siemens inverter [56]. Due to the use of a highly dynamic servo motor, the test setup is an axis with dominating mechanics, whose first position-dependent natural frequency (61–66 Hz) limits the bandwidth of the position control loop. This is the first axial eigenmode. The speed control loop is tuned to provide a damping $D \approx 0.5$, which is a common industrial practice for feed drives, as a compromise between good disturbance damping and small tracking errors. This resulted in $K_{\mathrm{p,n}} = 175$ 1/s, $K_{\mathrm{i,n}} = 50$ 1/s and in the $PT_2I$-parameters $\omega_0 = 277.8$ rad/s and $D = 0.518$ of the closed loop dynamics of the speed control loop, which were determined via frequency response analysis. The plant approximation without integrator $G_P(s) = \mathcal{L}(\dot{x}_a)/\mathcal{L}(\dot{x}_s)$ is also visualized on Fig. 5, where the closed-loop speed-controlled plant was recorded at both ends of the axis ($x = 0$ m and $x = 0.6$ m) with and without an additional load mass of $m_l \approx 110$ kg. Although, the mechanical resonance frequencies lie between 51 Hz ($x = 0.6$ m, $m_l = 110$ kg) and 66 Hz ($x = 0$ m, $m_l = 0$ kg), it can be seen on Fig. 5, that closing the PI velocity control loop already captures most of the variations. This leads to plant variations for the position controller of $\Delta\omega_0 \in [-10, 0]$ % and $\Delta D \in [0, 20]$ %. The plant approximation only roughly captures the model behavior: The static gain $G_P(s \to 0)$ does not match and the dynamics for frequencies > 100 Hz deviate with more than 10 dB. The following evaluation sections, however, will show that capturing the approximate model dynamics already leads to improvements in the controller performance and that the controller is robust enough to cope with the model deviations.

The actual acceleration of the machine table is reconstructed with the help of a Kalman filter [54], which uses the system model according to (21) for the state estimation. The filter is designed by specifying constant covariance matrices for the stochastic noise processes. While the covariance matrix for the measurement noise $\boldsymbol{R}$ of the table-mounted direct measurement system is determined offline by noise measurements, the process noise $\boldsymbol{Q}$ is determined empirically by an iterative approach. For both noise processes, it is also assumed that the individual noise variables are stochastically independent, resulting in the diagonal matrices listed in Table 2.

To evaluate the controller's performance and for a comparison with the state-of-the-industry P-PI controller, their respective behavior is evaluated in terms of command tracking and disturbance rejection. Both aspects are demonstrated in time and frequency domain. This leads to four excitation trajectories, which are parameterized as described in Table 3. Command tracking behavior in time domain is shown with a time-optimal ramp trajectory smoothed by an s-shaped profile of order four [57] (cf. Fig. 6), i.e. with limited dynamics of the derivatives up to jerk level. It is industrial standard to smooth the setpoints by this or a similar profile. To analyze the transient behavior, a highly dynamic motion is deliberately aimed for, since the controller is most challenged by the large setpoint changes during the acceleration

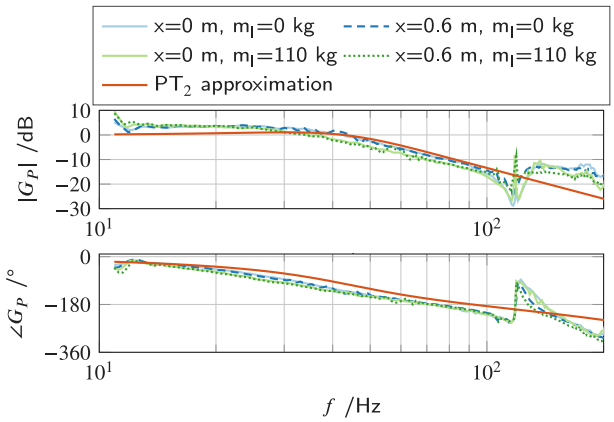



**Fig. 5.** Approximation of the closed-loop speed-controlled plant $G_P$ ($\dot{x}_s \to \dot{x}_a$) with a $PT_2$ system. Numeric transfer functions are recorded for table positions $x$ at both ends of the spindle and for varying load masses $m_l$.

**Table 1**
Basic information of the test bench.

| Module/parameter | Type/value |
|---|---|
| Servo drive | Siemens 1FT7085-7WF71-1ML1 |
| Max. torque | 43 Nm |
| Direct drive | Bosch Rexroth LSP200U-RU2-BN |
| Max. force | 13.2 kN |
| Ball screw | Steinmeyer 3526/40.40.1800.2000EP3 |
| Pitch | 40 mm |
| Diameter | 40 mm |
| Nut | Steinmeyer 3526/40.40.6.3 Eta+ |
| Incremental encoder | Heidenhain LS 187C |
| Grating period | 20 µm |
| Traversing velocity | 2000 mm/s |
| Upsampling factor | 8192 |
| Machine table mass | 400 kg |

**Table 2**
Covariance matrices of the Kalman filter.

| Noise covariance | Parameter |
|---|---|
| Process noise | $\boldsymbol{Q} = 10^{-6}\ \mathrm{mm}^2 \cdot \mathrm{diag}\{1,\ 1/\mathrm{s}^2,\ 100/\mathrm{s}^4\}$ |
| Measurement noise | $\boldsymbol{R} = 10^{-2}\ \mathrm{mm}^2 \cdot \mathrm{diag}\{1,\ 10/\mathrm{s}^2\}$ |

and deceleration process. In frequency domain, the transfer function of command tracking behavior is generated with a pseudorandom binary sequence (PRBS)

$$x_{\mathrm{s,P}}(t) = \begin{cases} \hat{x}_{\mathrm{P}} + \dot{x}_{\mathrm{s,P}} t, & \text{if } \zeta(t, f_{\mathrm{max,P}}) > 0 \\ -\hat{x}_{\mathrm{P}} + \dot{x}_{\mathrm{s,P}} t, & \text{otherwise,} \end{cases} \tag{31}$$

with amplitude $\hat{x}_{\mathrm{P}}$ where $\zeta$ is a uniform, zero-mean random number generator only being evaluated with frequency $f_{\mathrm{max,P}}$ and $\dot{x}_{\mathrm{s,P}}$ is an offset to prevent nonlinear stick–slip. To measure the disturbance rejection behavior disturbance forces are applied to the table of the ball screw drive with a force-controlled linear direct drive, hence directly influencing the actual table position $x_a$. During the disturbance force measurements, the desired position is controlled at $x_{\mathrm{s}} \equiv 0$ mm. In time domain, a step signal

$$z_\sigma(t) = \hat{F}_\sigma \sigma(t) \tag{32}$$

with step height $\hat{F}_\sigma$ and the Heaviside step function $\sigma(t)$ is selected. In frequency domain, the input force is given by a linear chirp signal

$$z_{\mathrm{c}}(t) = \hat{F}_{\mathrm{c}} \sin\left(2\pi\left(f_{\mathrm{min,c}} + \frac{f_{\mathrm{max,c}} - f_{\mathrm{min,c}}}{T_{\mathrm{c}}} t\right)\frac{t}{T_{\mathrm{c}}}\right), \tag{33}$$

which excites all frequencies between $f_{\mathrm{min,c}}$ and $f_{\mathrm{max,c}}$ with a force amplitude $\hat{F}_{\mathrm{c}}$ in the time interval $t \in [0, T_{\mathrm{c}}]$.

**Table 3**
Feed parameters of the reference inputs.

| | Time domain | | Frequency domain | |
|---|---|---|---|---|
| | Ramp trajectory | | PRBS noise | |
| Command tracking | $\dot{x}_{\mathrm{s}}$ | ≤750 mm/s | $\hat{x}_{\mathrm{P}}$ | 0.1 mm |
| | $\lvert\ddot{x}_{\mathrm{s}}\rvert$ | $\leq 8 \cdot 10^3$ mm/s$^2$ | $f_{\mathrm{max,P}}$ | 400 Hz |
| | $\lvert\dddot{x}_{\mathrm{s}}\rvert$ | $\leq 1 \cdot 10^6$ mm/s$^3$ | $\dot{x}_{\mathrm{s,P}}$ | 8 mm/s |
| | Disturbance step | | Disturbance chirp | |
| Disturbance rejection | $\hat{F}_\sigma$ | 500 N | $\hat{F}_{\mathrm{c}}$ | 150 N |
| | | | $f_{\mathrm{min,c}}$ | 0.1 Hz |
| | | | $f_{\mathrm{max,c}}$ | 100 Hz |
| | | | $T_{\mathrm{c}}$ | 16.25 s |

### 6.1. Validation and control parameter analysis

The validation of the new SMC-PI controller is carried out for the reference inputs described in the previous section. For a direct comparison of the performance the controller parameter $\lambda$ is varied. The following results are shown for choices of $K \equiv \lambda$, as further experimental investigations confirmed that choosing $K$ different from $\lambda$ does not result in significantly changed tracking or disturbance behavior. The controller gain is chosen as $\lambda \in 10^2 \cdot \{1.5, 2, 2.5, 3, 3.5\}$ 1/s. Values of $\lambda \geq 400$ 1/s showed unstable behavior on the test bench for the demanding parameter values of the ramp trajectory due to oscillations presumably caused by amplification of plant effects that are not part of the internal $PT_2I$ model (21).

The tracking error for the reference ramp trajectory is shown in Fig. 7a for a varying controller gain, the tracking behavior for $\lambda = 300$ 1/s is further shown in Fig. 6 in comparison with the reference trajectory. As expected, the tracking error is reduced by increasing the controller gain. The start-up and braking acceleration phases of the machine table can be seen, which are indicated by the minimum and maximum deflections of the control error at the beginning and end of the movement. These deviations are mainly caused by the high dynamics in combination with the dead time resulting from the communication between the rapid prototyping system and the inverter and can be expected to be reduced e.g., by directly integrating the controller into the inverter. For the constant velocity phase $t \in [0.18, 0.27]$ s the tracking error converges to 0 mm, where a slight oscillation motion remains almost independently of the selected parameterization. Further, the tracking error is aperiodically damped for all shown controller gain values as it decays without recognizable overshoot. The command tracking behavior in frequency domain is shown in Fig. 7b, where the open loop transfer function $G_0(\mathrm{i}\omega) = E_x(\mathrm{i}\omega)/X_a(\mathrm{i}\omega)$ with the Fourier transformed tracking error $E_x(\mathrm{i}\omega)$ and actual position value $X_a(\mathrm{i}\omega)$. For small frequencies the integral behavior of the plant dominates with $|G_0|$ falling with 20 dB/decade and $\angle G_0$ starting at −90°. This leads to smaller tracking errors. Higher controller gains lift the amplitude $|G_0|$ and increase the gain crossover frequency $\omega_{0\mathrm{dB}}$, where $|G_0|$ crosses the 0 dB line, from 4.6 Hz ($\lambda = 150$ 1/s) to 19.7 Hz ($\lambda = 350$ 1/s). As a direct measure of the convergence velocity, the increasing gain crossover frequency affirms the tracking results in the time domain. Further, the increase of the controller gain also lifts the phase leading to nearly invariant phase reserves and thus to constant robustness properties, almost independent of the chosen controller gain. For higher frequencies, where the internal $PT_2I$ model differs from the plant behavior, the uncompensated dynamics are amplified as well by increasing the controller gain. For $\lambda = 350$ 1/s the resonance gain at 135 Hz raises close to 0 dB. Hence, during the parameterization attention must be paid to unmodeled plant dynamics, which set an upper limit for the achievable controller gain.

The force step response in Fig. 8a shows a disturbance rejection response similar to the command tracking results. The position error caused by the disturbance force is compensated, whereby an increasing controller gain reduces the tracking error. The disturbance force step

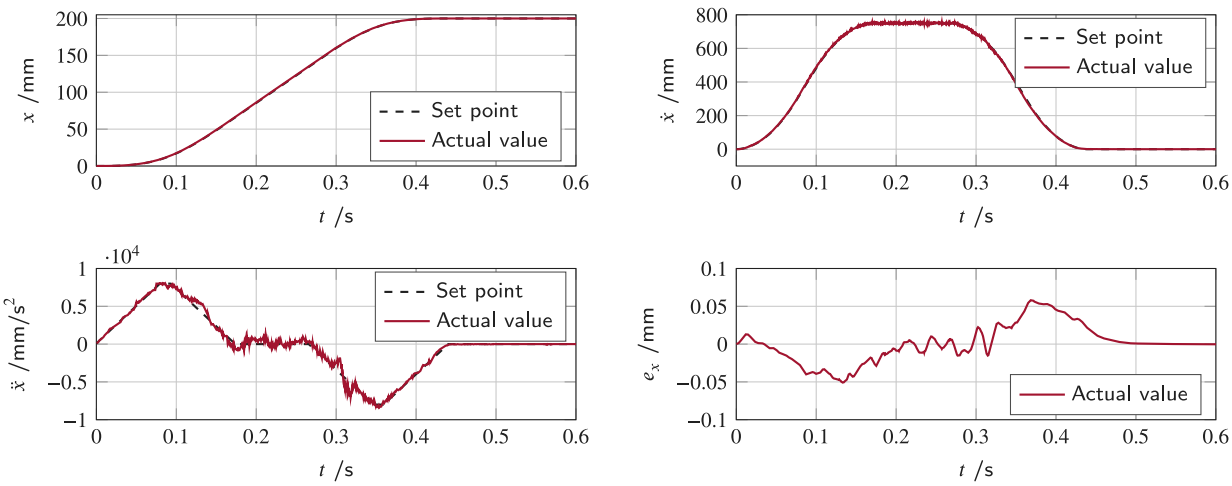
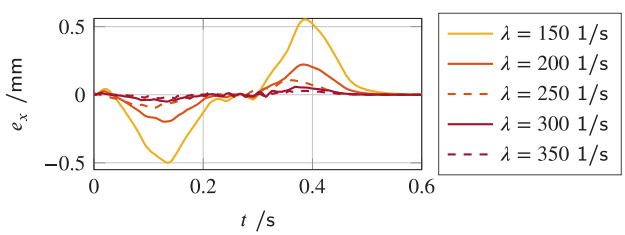
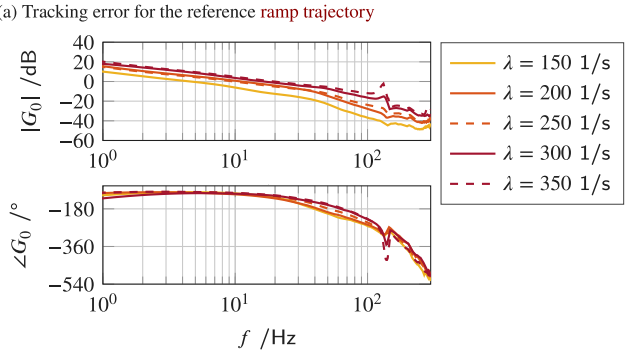



**Fig. 6.** Position, velocity, acceleration and tracking error for SMC-PI cascade control with $\lambda_a = 300$ 1/s for the reference ramp trajectory.

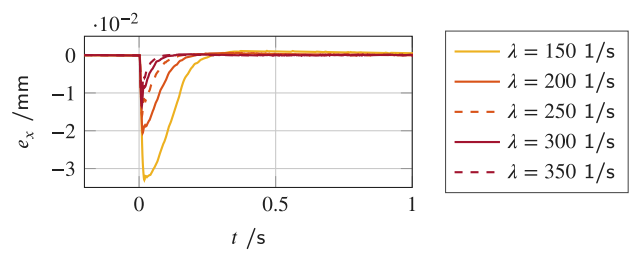



(a) Tracking error for the reference ramp trajectory

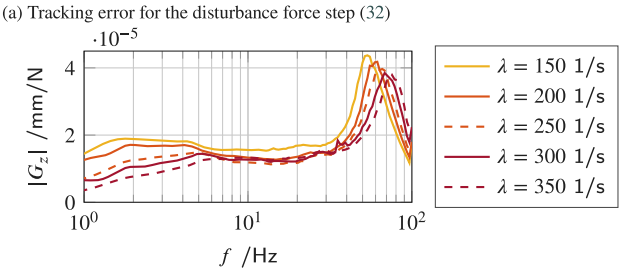



(b) Command frequency response for the reference PRBS input (31)

**Fig. 7.** Setpoint tracking behavior in time and frequency domain for different settings for $\lambda$.




(a) Tracking error for the disturbance force step (32)




(b) Compliance frequency response for the chirp signal (33)

**Fig. 8.** Disturbance rejection behavior in time and frequency domain for different settings for $\lambda$.

is compensated aperiodically damped, which is important for applications requiring steady tracking errors under disturbance forces, as milling. Only $\lambda = 150$ 1/s shows a minor overshoot indicating that this controller gain is chosen slightly too low for good disturbance rejection. The amplitude of the compliance transfer function $G_z(\mathrm{i}\omega) = X_a(\mathrm{i}\omega)/Z(\mathrm{i}\omega)$ with the Fourier transformed actual position $X_a(\mathrm{i}\omega)$ and disturbance force $Z(\mathrm{i}\omega)$ is shown in Fig. 8b. The SMC behaves fundamentally different from a P position controller, which usually linearly amplifies the resonance amplitude of $G_z(\mathrm{i}\omega)$: An increasing controller gain not only dampens the amplitude of the resonance from 0.043 to 0.037 µm/N and also shifts the resonance frequency from 53.5 Hz to 75.8 Hz. Lower frequencies are damped as well with an increasing controller gain. An exception is again $\lambda = 150$ 1/s, which also shows a higher compliance in mid-frequency ranges around 10 Hz.

Overall, the proposed SMC-PI shows good command tracking and disturbance rejection without overshooting in the tracking error, stationary errors converging to zero and being stable over a wide range of settings for the controller parameters, despite the chosen $PT_2I$ model, which only captures the essential dynamics of the plant. As the smallest controller parameter dominates the overall dynamics, $K \equiv \lambda$ was used and can be recommended to reduce complexity of the parameterization. A good range for the controller parameters at the test bench used was found to lie within $\lambda \in [200, 300]$ 1/s, since lower gains show disproportionately worse disturbance rejection (Fig. 8b) and higher gains excite the unmodeled plant dynamics in a range, where the amplitudes grow above 0 dB (Fig. 7b). Thus, for further comparison and robustness analysis the boundaries $\lambda = 200$ 1/s and $\lambda = 300$ 1/s of the interval are selected.

### 6.2. Comparison with the P-PI cascade control

The proposed SMC-PI is now compared with the state-of-the-industry P-PI controller. For a fair comparison, the P-PI controller is parameterized with velocity feedforward control, i.e. the desired velocity is directly added to the input of the speed control loop. The P gain $K_v$ is then parameterized to have a stability margin of $k_r > 10$ dB comparable to the SMC-PI (see Table 4). This method yields in $K_v = 50$ 1/s, which is at $\approx 30\,\%$ of the critical gain ($K_{v,c} = 168$ 1/s) also the maximum possible gain for aperiodic damping of a step response on the test bench without feedforward [58].

**Table 4**
Performance of the respective controllers.

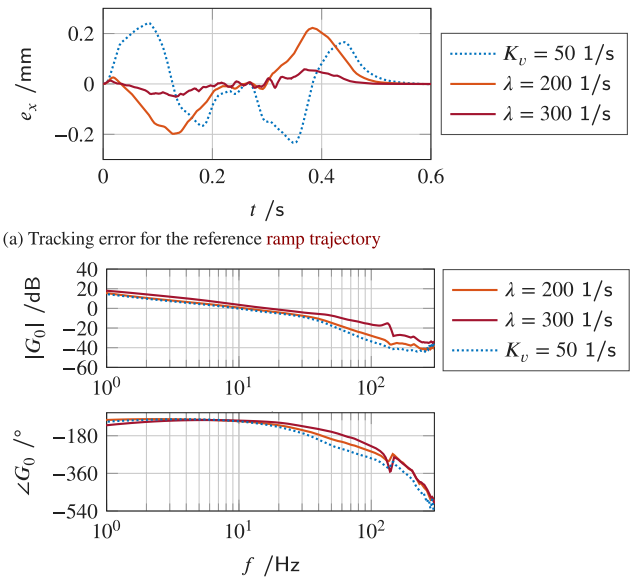
| Controller | P-PI | SMC-PI | |
|---|---|---|---|
| | $K_v = 50$ 1/s | $\lambda = 200$ 1/s | $\lambda = 300$ 1/s |
| Time domain | | | |
| $J_1(e)$/mm | 0.0986 | 0.0722 | 0.0174 |
| std($e$)/mm | 0.1245 | 0.1024 | 0.0244 |
| max ($e$)/mm | 0.2426 | 0.2225 | 0.0585 |
| Frequency domain | | | |
| $k_r$/dB | 10.19 | 11.40 | 10.54 |
| $f_{-180°}$/Hz | 34.17 | 42.11 | 62.00 |
| $\phi_r$/° | 69.49 | 67.61 | 66.75 |
| $f_{0dB}$/Hz | 9.70 | 11.06 | 16.40 |

The controllers are benchmarked with the same reference inputs, which are defined in Section 6 and parameterized according to Table 3. For a qualitative comparison between the controllers the tracking error for the ramp trajectory is evaluated with the mean absolute error functional $J_1(e) = \frac{1}{N}\sum_{i=1}^{N}|e_i|$, the maximum tracking error and the standard deviation $\mathrm{std}(e) = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}|e_i - \frac{1}{N}\sum_{i=1}^{N} e_i|^2}$ with the number of sampling points $N$. In frequency domain the tracking behavior is further quantified by the gain margin $k_r$, i.e. the maximum gain that can be applied to the transfer function before the amplitude becomes unstable on the negative real axis $f_{-180°}$, and the phase margin $\phi_r$, which is the angle between the transfer function value of the crossover frequency $f_{0dB}$ and the critical point. Both, the performance measures in time and frequency domain are shown in Table 4.

The command tracking results in time domain, shown in Fig. 9 and the performance results in Table 4 show a quantitatively similar performance of the P-PI controller and the SMC-PI with $\lambda = 200$ 1/s. However, for the P-PI controller this is the upper gain limit, since higher gains leading to overshoot of the tracking error. For the SMC-PI $\lambda = 200$ 1/s is the lower bound on control gains and by increasing $\lambda$ the tracking error can be drastically reduced by > 75% in all error measures. The results are confirmed in the frequency domain (see Fig. 9b), where the amplitudes of the SMC-PI with $\lambda = 200$ 1/s and the P-PI are equivalent up to $\omega_0$. For $\lambda = 300$ 1/s the crossover frequency is increased to 16.4 Hz, which explains the much lower tracking errors. At the same time, amplitude and phase margins are insensitive to changes in $\lambda$ and remain comparable to the P-PI controller for the chosen SMC gains.

For the disturbance rejection behavior, which is shown on Fig. 10 in time and frequency domain, the controllers show significant differences. The maximum positioning error for a disturbance step of 500 N on Fig. 10a is reduced from 32.1 µm for the P-PI controller to 20.6 µm ($\lambda = 200$ 1/s) and 13.5 µm ($\lambda = 300$ 1/s) due to the higher order derivatives being considered in the SMC. The P-PI controller even shows a slight overshoot for times around 0.5 s indicating, that no further increase in the $K_v$ gain can be achieved without worsening the aperiodic damping. In the compliance frequency response on Fig. 10b the better disturbance rejection can also be seen, where the SMC-PI not only shows lower compliance over the whole frequency range, but also shifts the resonance peak to higher frequencies, which is even amplified by increasing $\lambda$.

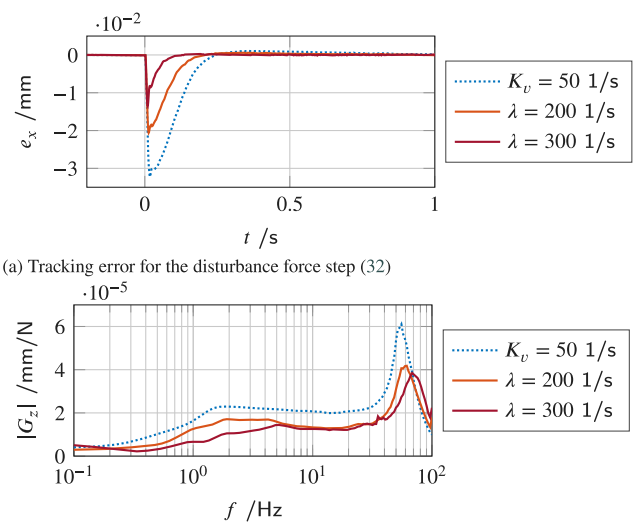
To evaluate the proposed controller under machining conditions, milling forces were simulated in closed loop based on the actual table velocity. The milling forces were generated with the axial component of the 2-DOF milling force model from [59], which was parameterized for HSC of low strength steel and a carbide cutter (see Table 5) and applied with the linear direct drive. The ball screw trajectory during milling is an s shaped profile ($\dot{x}_s \leq 500$ mm/s, $\dot{x}_s/\ddot{x}_s$ limited as in Table 3) moving the table over the whole axis length of 700 mm. Note, that we chose peripheral milling in contrast to face milling due to its higher cutting forces. Fig. 11a shows the applied cutting force during the constant velocity phase, the resulting position error for P-PI and SMC are shown on Fig. 11b and numerically evaluated in Table 6. The




(a) Tracking error for the reference ramp trajectory

(b) Command frequency response for the reference PRBS input (31)

**Fig. 9.** Setpoint tracking behavior in time and frequency domain as comparison between SMC with $\lambda = \{200, 300\}$ 1/s and the P-PI controller with velocity feedforward and $K_v = 50$ 1/s.



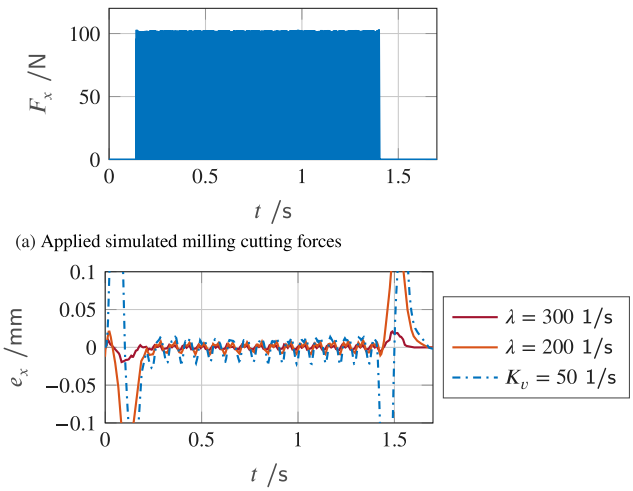

(a) Tracking error for the disturbance force step (32)

(b) Compliance frequency response for the chirp signal (33)

**Fig. 10.** Disturbance rejection behavior in time and frequency domain as comparison between SMC with $\lambda = \{200, 300\}$ 1/s and the P-PI controller with velocity feedforward and $K_v = 50$ 1/s.

**Table 5**
Parameters for the HSC milling validation experiment.

| Parameter | Value | Unit |
|---|---|---|
| Cutting speed $v_c$ | 1580 | m/min |
| Spindle speed $n$ | 25 150 | rpm |
| Spindle feed $\dot{x}_s$ | 30 | m/min |
| Infeed $a_p$ | 0.5 | mm |
| Width of cut $a_e$ | 2 | mm |
| Tool diameter $d$ | 20 | mm |
| Number of teeth $N_t$ | 4 | – |
| Specific cutting force $k_c$ | 1717 | N/mm$^2$ |
| Specific radial force $k_n$ | 928 | N/mm$^2$ |
| Material constant cutting $m_c$ | 0.23 | – |
| Radial material constant $m_n$ | 0.39 | – |




(a) Applied simulated milling cutting forces

(b) Tracking error for the controllers

**Fig. 11.** Setpoint tracking behavior for simulated HSC milling experiment with parameters from Table 5.

**Table 6**
Tracking performance for simulated HSC milling forces.

| Controller | P-PI | SMC-PI | |
|---|---|---|---|
| | $K_v = 50$ 1/s | $\lambda = 200$ 1/s | $\lambda = 300$ 1/s |
| $J_1(e)$/mm | 0.0394 | 0.0194 | 0.0036 |
| std($e$)/mm | 0.0742 | 0.0409 | 0.0056 |
| max ($e$)/mm | 0.2496 | 0.1438 | 0.0211 |

results are comparable to the disturbance rejection experiments: The SMC results in lower maximum errors during the acceleration phases ($t \in [0, 0.14] \cup [1.38, 1.52]$ s), where the error amplitude is cut by 42.4% for the SMC-PI with $\lambda = 200$ 1/s compared to P-PI control and even further with $\lambda = 300$ 1/s by 91.6%. The mean absolute error $J_1$ is reduced in a similar order.

Overall, the novel SMC-PI shows a significant potential in comparison to the state of the art that is nowadays applied in industry for both command tracking and disturbance rejection for a wide range of the controller gain $\lambda$ (see Table 6).

### 6.3. Analysis of the robustness

In addition to the investigation of the performance and as a continuation of the analytical uncertainty analysis in Section 5.1, experimental robustness studies were carried out, which are separated in a classical linear robustness analysis based on the Nyquist diagram of the open loop transfer function and a robustness against wrongly chosen model parameters.

A sufficiently high stability margin is a prerequisite for good robustness against deviations of the system behavior from the nominal behavior. Hence, a gain margin $k_r > 8$ dB and a phase margin $\phi_r > 60°$ are usually required for feed drives. Fig. 12 illustrates the Nyquist plot of the open position control loop for the controllers already considered, generated with the PRBS noise signal from Table 3. The curves and Table 4 show that the requirements are met by all controllers and that the stability margins are very similar to each other, but the frequencies belonging to the respective margins are much higher for SMC-PI. This indicates comparable robustness characteristics for the selected parameterizations. It is remarkable that the SMC-PI shows only slight changes in the stability margin, despite the large difference in parameters. This is a consequence of the aperiodically damped controller behavior and the lifted phase in the command frequency response.

As the control law (22) is model-based, the resiliency to errors in the model parameter identification is shown. The tracking error of the

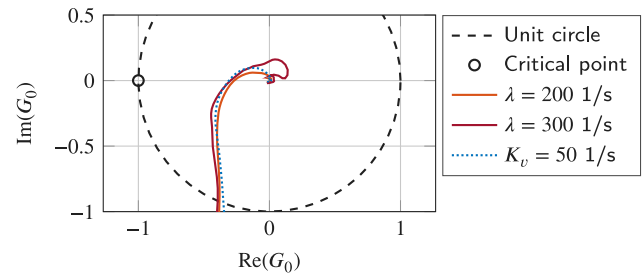
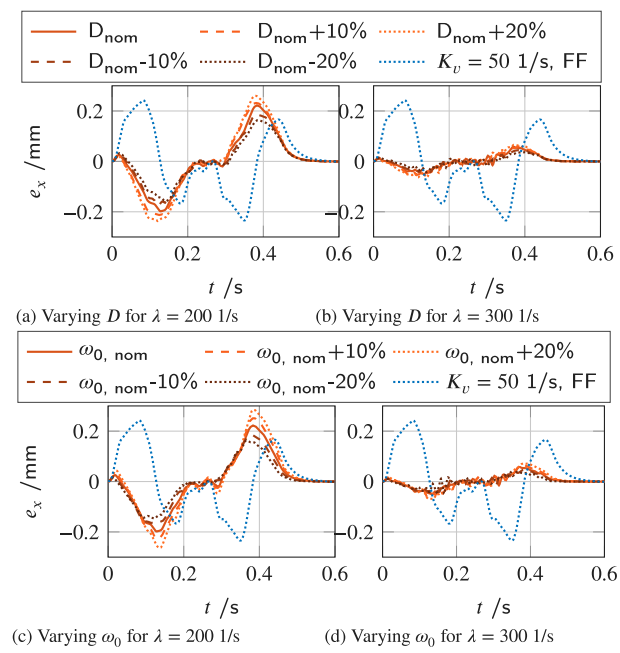



**Fig. 12.** Nyquist plot of open loop tracking response as comparison between SMC with $\lambda = \{200, 300\}$ 1/s and the P-PI controller with velocity feedforward and $K_v = 50$ 1/s.




(a) Varying $D$ for $\lambda = 200$ 1/s (b) Varying $D$ for $\lambda = 300$ 1/s

(c) Varying $\omega_0$ for $\lambda = 200$ 1/s (d) Varying $\omega_0$ for $\lambda = 300$ 1/s

**Fig. 13.** Robustness of the tracking error against variations of $D$ and $\omega_0$ within ±20 % for $\lambda = \{200, 300\}$ 1/s, the tracking error of the P-PI controller is plotted for comparison.

reference ramp trajectory is analyzed by varying the $PT_2$ parameters. For both ends of the spindle and an additional load mass of 110 kg the effect on the plant parameters change less than 20% (cf. Fig. 5), hence a variation of $\omega_0$ and $D$ by ±20 % around their nominal values $\omega_{0,\text{nom}} = 277.8$ rad/s and $D_{\text{nom}} = 0.518$ is chosen for SMC gains of $\lambda = 200$ 1/s and $\lambda = 300$ 1/s. As the plant behaves basically linear, this variation is equivalent to keeping the controller parameters and modifying the plant, e.g. by varying the load mass $m_l$. Measurements on the test bench shown on Fig. 13 yield a stable tracking behavior for both controller gains and all variations. As expected, the tracking error is less sensitive against a variation of $D$ than a variation of $\omega_0$, as $D$ only enters the control term as part of the compensation (23), whereas $\omega_0$ is a normalization factor of the input gain and directly influences all parts of the control law. Larger values of $\omega_0$ weaken the control input $u$ and lead to larger tracking errors, while smaller values result in a more aggressive controller. Even with parameter variations of ±20 % the novel SMC-PI shows tracking errors comparable to P-PI control for $\lambda = 200$ 1/s and far less tracking errors for $\lambda = 300$ 1/s. The robustness of the controller against mismatch model parameters is an inherent property of the design method based on Lyapunov stability and the two degrees-of-freedom approach, where the model compensation is independent of the feedback terms.

Thus, the SMC-PI is robust far beyond the expected uncertainties in the model parameters $\omega_0$ and $D$ and shows decreased tracking errors for $\lambda = 300$ 1/s compared to the P-PI despite model errors.

### 6.4. Discussion of the findings

Due to the robustness against model errors, the identification of the model parameters $\omega_0$ and $D$ is not critical. The knowledge about these parameters results, explicitly or at least implicitly, from the parameterization of the speed control loop. The control parameter $\lambda$ is non-linearly related to the amplitude and phase margin. Therefore, it is not sufficient to determine $k_r$ once with an initial $\lambda_0$ and to calculate the optimum value for $\lambda$ analytically from it. So far, the optimum has to be found iteratively. However, it can be stated that $\lambda$ should be set as high as the stability allows, since this will improve both the command tracking behavior and the disturbance rejection holistically. This is worth mentioning because it is in contrast to P-PI cascade control, where higher $K_v$ increases compliance in the region of the first natural frequency.

Regarding applications, only machines with direct measuring system are suitable. This narrows down the applications to high-precision processes, as for lower accuracy requirements position control is often realized purely via the motor encoders. The high robustness against system parameter variations allows the use for parameter-varying processes, e.g., material-removing and material-applying processes. In addition, the SMC-PI is suitable for applications with process forces (e.g. milling, turning, etc.) due to its increased disturbance rejection behavior. Just as for high-speed applications with the requirement for low tracking errors. Taking all these properties into account, the SMC-PI is particularly suited for HSC, but also for many other applications.

## 7. Conclusions

In this paper, a sliding mode position control concept (SMC-PI cascade control) for improving the dynamic behavior of elastic feed drives has been introduced. The SMC-PI was designed to bridge the gap between cascaded control with a P position controller, which is current industrial practice due to its easy parameterization and already good robustness and tracking performance, and higher control approaches, which are harder to parameterize and lower industrial acceptance. The control law is derived on a general linear system in controllable canonical form based on the Lyapunov stability condition and applied to an elementary elastic feed drive model as direct replacement for the position controller with the two simply identifiable system parameters $\omega_0$ and $D$, which are already known from the parameterization of the speed control loop. Uncertainties in the model parameters are explicitly analyzed and the SMC is shown to be stable for model mismatch in $\omega_0$, well damped for a wide range of more than $\pm 50$ % and insensitive against variations in $D$. The controller parameters directly modulate the error dynamics of the closed loop system and can be reduced to only one gain factor ($K \equiv \lambda$), which leads to a parameterization effort for a commissioning engineer comparable to the state-of-the-art P position controller.

The controller design was experimentally evaluated and validated regarding its parameterization, its linear robustness properties and robustness against model uncertainties as well as compared to a P controller with velocity feedforward control. Compared to the conventional P-position control loop the proposed approach shows improved tracking performance and enhanced disturbance rejection, whereby the maximum tracking error of a demanding ramp trajectory could be reduced by 75.9% and by 57.9% for a disturbance step of 500 N on the test bench. An additional HSC experiment with simulated milling forces being applied to the feed drive confirms the results with a reduced tracking error during milling of 82 %. In addition, excellent robustness properties were demonstrated for stability margins to be almost unchanging over a wide variation of the new controller's parameterization and against variations of the model parameters in the range of $\pm 20$ %.

Finally, the presented extension of the position controller with an SMC is not limited to the specific setup and the gained insight can be applied more generally: the experimental results were obtained with the industrial setup of a machine axis, such that comparable results can be expected from the integration in a machine tool. With the focus on applicability, the velocity and current cascades can be kept on integration and only one controller gain needs to be tuned. Moreover, by using a general compliant drive model, the presented approach is not limited to the application on ball-screw drives, but is expected to work well also with other feed drive classes under the precondition that they provide a direct measurement of the load side.

## CRediT authorship contribution statement

**Michael Neubauer:** Conceptualization, Methodology, Validation, Investigation, Writing - original draft, Writing - review & editing, Visualization. **Felix Brenner:** Conceptualization, Methodology, Validation, Investigation, Writing - original draft, Writing - review & editing, Visualization. **Christoph Hinze:** Conceptualization, Methodology, Validation, Investigation, Writing - original draft, Writing - review & editing, Visualization. **Alexander Verl:** Supervision, Project administration, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## References

[1] Y. Altintas, A. Verl, C. Brecher, L. Uriarte, G. Pritschow, Machine tool feed drives, CIRP Ann. 60 (2) (2011) 779–796.
[2] G.-H. Feng, Y.-L. Pan, Investigation of ball screw preload variation based on dynamic modeling of a preload adjustable feed-drive system and spectrum analysis of ball-nuts sensed vibration signals, Int. J. Mach. Tools Manuf. 52 (1) (2012) 85–96, http://dx.doi.org/10.1016/j.ijmachtools.2011.09.008.
[3] G. Pritschow, N. Croon, Ball screw drives with enhanced bandwidth by modification of the axial bearing, CIRP Ann. 62 (1) (2013) 383–386, http://dx.doi.org/10.1016/j.cirp.2013.03.086.
[4] M. Wang, T. Zan, X. Gao, S. Li, Suppression of the time-varying vibration of ball screws induced from the continuous movement of the nut using multiple tuned mass dampers, Int. J. Mach. Tools Manuf. 107 (2016) 41–49, http://dx.doi.org/10.1016/j.ijmachtools.2016.05.003.
[5] K. Erkorkmaz, Y. Altintas, High speed CNC system design. Part I: jerk limited trajectory generation and quintic spline interpolation, Int. J. Mach. Tools Manuf. 41 (9) (2001) 1323–1345, http://dx.doi.org/10.1016/s0890-6955(01)00002-5.
[6] N.C. Singer, W.P. Seering, Preshaping command inputs to reduce system vibration, J. Dyn. Syst. Meas. Control 112 (1) (1990) 76–82, http://dx.doi.org/10.1115/1.2894142.
[7] H.-W. Huang, M.-S. Tsai, Y.-C. Huang, Modeling and elastic deformation compensation of flexural feed drive system, Int. J. Mach. Tools Manuf. 132 (2018) 96–112, http://dx.doi.org/10.1016/j.ijmachtools.2018.05.002.
[8] K. Zhang, A. Yuen, Y. Altintas, Pre-compensation of contour errors in five-axis CNC machine tools, Int. J. Mach. Tools Manuf. 74 (2013) 1–11, http://dx.doi.org/10.1016/j.ijmachtools.2013.07.003.
[9] K. Erkorkmaz, Y. Altintas, High speed CNC system design. Part III: high speed tracking and contouring control of feed drives, Int. J. Mach. Tools Manuf. 41 (11) (2001) 1637–1658, http://dx.doi.org/10.1016/S0890-6955(01)00004-9.
[10] D.J. Gordon, K. Erkorkmaz, Accurate control of ball screw drives using pole-placement vibration damping and a novel trajectory prefilter, Precis. Eng. 37 (2) (2013) 308–322, http://dx.doi.org/10.1016/j.precisioneng.2012.09.009.
[11] D. Sepasi, R. Nagamune, F. Sassani, Tracking control of flexible ball screw drives with runout effect and mass variation, IEEE Trans. Ind. Electron. 59 (2) (2012) 1248–1256, http://dx.doi.org/10.1109/tie.2011.2158042.
[12] N. Uchiyama, Adaptive two-degree-of-freedom control of feed drive systems, Int. J. Mach. Tools Manuf. 48 (3) (2008) 437–445, http://dx.doi.org/10.1016/j.ijmachtools.2007.08.025.
[13] C. Zhang, Y. Chen, Tracking control of ball screw drives using ADRC and equivalent-error-model-based feedforward control, IEEE Trans. Ind. Electron. 63 (12) (2016) 7682–7692, http://dx.doi.org/10.1109/TIE.2016.2590992.

[14] D. Papageorgiou, M. Blanke, H.H. Niemann, J.H. Richter, Friction-resilient position control for machine tools—Adaptive and sliding-mode methods compared, Control Eng. Pract. 75 (2018) 69–85, http://dx.doi.org/10.1016/j.conengprac.2018.03.017.

[15] J. Guldner, V.I. Utkin, The chattering problem in sliding mode systems, in: Fourteenth Intenational Symposium of Mathematical Theory of Networks and Systems, MTNS2000, 2000.

[16] H. Lee, Chattering Suppression in Sliding Modecontrol System (Ph.D. thesis), Ohio State University, 2007.

[17] M.-L. Tseng, M.-S. Chen, Chattering reduction of sliding mode control by low-pass filtering the control signal, Asian J. Control 12 (3) (2010) 392–398, http://dx.doi.org/10.1002/asjc.195.

[18] Y. Altintas, K. Erkorkmaz, W.-H. Zhu, Sliding mode controller design for high speed feed drives, CIRP Ann. 49 (1) (2000) 265–270, http://dx.doi.org/10.1016/s0007-8506(07)62943-6.

[19] K. Erkorkmaz, A. Kamalzadeh, High bandwidth control of ball screw drives, CIRP Ann. 55 (1) (2006) 393–398, http://dx.doi.org/10.1016/S0007-8506(07)60443-0.

[20] A. Kamalzadeh, K. Erkorkmaz, Compensation of axial vibrations in ball screw drives, CIRP Ann. 56 (1) (2007) 373–378, http://dx.doi.org/10.1016/j.cirp.2007.05.087.

[21] Y. Altintas, C.E. Okwudire, Dynamic stiffness enhancement of direct-driven machine tools using sliding mode control with disturbance recovery, CIRP Ann. 58 (1) (2009) 335–338, http://dx.doi.org/10.1016/j.cirp.2009.03.045.

[22] A. Kamalzadeh, K. Erkorkmaz, Accurate tracking controller design for high-speed drives, Int. J. Mach. Tools Manuf. 47 (9) (2007) 1393–1400, http://dx.doi.org/10.1016/j.ijmachtools.2006.08.027.

[23] C. Okwudire, Y. Altintas, Minimum tracking error control of flexible ball screw drives using a discrete-time sliding mode controller, CIRP Ann. 131 (5) (2009) 367, http://dx.doi.org/10.1115/1.3155005.

[24] A.E.K. Mohammad, N. Uchiyama, S. Sano, Reduction of electrical energy consumed by feed-drive systems using sliding-mode control with a nonlinear sliding surface, IEEE Trans. Ind. Electron. 61 (6) (2014) 2875–2882, http://dx.doi.org/10.1109/TIE.2013.2275975.

[25] H.-T. Yau, J.-J. Yan, Adaptive sliding mode control of a high-precision ball-screw-driven stage, Nonlinear Anal. RWA 10 (3) (2009) 1480–1489, http://dx.doi.org/10.1016/j.nonrwa.2008.01.025.

[26] K.R. Muske, H. Ashrafiuon, S. Nersesov, M. Nikkhah, Optimal sliding mode cascade control for stabilization of underactuated nonlinear systems, J. Dyn. Syst. Meas. Control 134 (2) (2012).

[27] G. Pritschow, C. Eppler, W.-D. Lehner, Ferraris sensor – the key for advanced dynamic drives, CIRP Ann. 52 (1) (2003) 289–292, http://dx.doi.org/10.1016/s0007-8506(07)60586-1.

[28] A. Verl, S. Frey, Improvement of feed drive dynamics by means of semi-active damping, CIRP Ann. 61 (1) (2012) 351–354, http://dx.doi.org/10.1016/j.cirp.2012.03.135.

[29] Z. Sun, G. Pritschow, A. Lechler, Enhancement of feed drive dynamics using additional table speed feedback, CIRP Ann. 65 (1) (2016) 357–360, http://dx.doi.org/10.1016/j.cirp.2016.04.099.

[30] Z. Sun, G. Pritschow, P. Zahn, A. Lechler, A novel cascade control principle for feed drives of machine tools, CIRP Ann. 67 (1) (2018) 389–392, http://dx.doi.org/10.1016/j.cirp.2018.03.004.

[31] D. Papageorgiou, M. Blanke, H.H. Niemann, J.H. Richter, Adaptive and sliding mode friction-resilient machine tool positioning – Cascaded control revisited, Mech. Syst. Signal Process. 132 (2019) 35–54, http://dx.doi.org/10.1016/j.ymssp.2019.06.015.

[32] A. Pisano, A. Davila, L. Fridman, E. Usai, Cascade control of PM DC drives via second-order sliding-mode technique, IEEE Trans. Ind. Electron. 55 (11) (2008) 3846–3854.

[33] H. Guo, Y. Liu, G. Liu, H. Li, Cascade control of a hydraulically driven 6-DOF parallel robot manipulator based on a sliding mode, Control Eng. Pract. 16 (9) (2008) 1055–1068.

[34] T. Tran, Q.P. Ha, H.T. Nguyen, Robust non-overshoot time responses using cascade sliding mode-pid control, J. Adv. Comput. Intell. Intell. Inform. (2007).

[35] M.P. Petronijević, Č. Milosavljević, B. Veselić, B. Peruničić-Draženović, S. Huseinbegović, Robust cascade control of electrical drives using discrete-time chattering-free sliding mode controllers with output saturation, Electr. Eng. (2021) 1–15, http://dx.doi.org/10.1007/s00202-020-01198-x.

[36] Y. Iwashita, T. Okita, Y. Toyozawa, K. Maeda, Servo control apparatus that performs dual-position feedback control, 2011, U.S. Patent 7915848B27.

[37] S. Kuenzel, T. Reichel, E. Schaefers, A. Uhlich, Method and apparatus for damping mechanical oscillations of a shaft in machine tools, manufacturing machines and robots, 2005, U.S. Patent 6903529B2.

[38] Dr. Johannes Heidenhain GmbH, Dynamic precision – Machining dynamically and with high accuracy technical information, 2013, URL: https://www.heidenhain.de/fileadmin/pdf/en/01_Products/Technische_Dokumentation/TI_Dynamic_Precision_ID1081196_en.pdf.

[39] F. Li, Y. Jiang, T. Li, K.F. Ehmann, Compensation of dynamic mechanical tracking errors in ball screw drives, Mechatronics 55 (2018) 27–37, http://dx.doi.org/10.1016/j.mechatronics.2018.08.004.

[40] S.I. Han, J.M. Lee, Adaptive dynamic surface control with sliding mode control and RWNN for robust positioning of a linear motion stage, Mechatronics 22 (2) (2012) 222–238, http://dx.doi.org/10.1016/j.mechatronics.2012.01.007.

[41] S.-C. Park, J.-M. Lee, S.-I. Han, Tracking error constrained terminal sliding mode control for ball-screw driven motion systems with state observer, Int. J. Precis. Eng. Manuf. 19 (3) (2018) 359–366, http://dx.doi.org/10.1007/s12541-018-0043-x.

[42] A.-C. Huang, Y.-S. Kuo, Sliding control of non-linear systems containing time-varying uncertainties with unknown bounds, Internat. J. Control 74 (3) (2001) 252–264, http://dx.doi.org/10.1080/00207170010003441.

[43] W. Lee, C.-Y. Lee, Y.H. Jeong, B.-K. Min, Distributed component friction model for precision control of a feed drive system, IEEE/ASME Trans. Mechatronics 20 (4) (2015) 1966–1974, http://dx.doi.org/10.1109/tmech.2014.2365958.

[44] D. Papageorgiou, M. Blanke, H.H. Niemann, J.H. Richter, Online friction parameter estimation for machine tools, Adv. Control Appl.: Eng. Ind. Syst. 2 (1) (2020) e28, http://dx.doi.org/10.1002/adc2.28.

[45] T. Hägglund, K.J. Åström, Revisiting the Ziegler-Nichols tuning rules for Pi control, Asian J. Control 4 (4) (2008) 364–380, http://dx.doi.org/10.1111/j.1934-6093.2002.tb00076.x.

[46] J.W. Umland, M. Safiuddin, Magnitude and symmetric optimum criterion for the design of linear control systems: what is it and how does it compare with the others? IEEE Trans. Ind. Appl. 26 (3) (1990) 489–497, http://dx.doi.org/10.1109/28.55967.

[47] Z. Preitl, Improving disturbance rejection by means of a double parameterization of the symmetrical optimum method, Sci. Bull. "Politeh." Univ. Timiş. Ser. Autom. Comput. (2005) 25–34.

[48] T. Berners, S. Kehne, A. Epple, C. Brecher, Hinf position control for machine tool feed drives, in: 7. WGP-Jahreskongress, 2017, pp. 391–399.

[49] X. Beudaert, O. Franco, K. Erkorkmaz, M. Zatarain, Feed drive control tuning considering machine dynamics and chatter stability, CIRP Ann. 69 (1) (2020) 345–348, http://dx.doi.org/10.1016/j.cirp.2020.04.054.

[50] S.M. Raafat, R. Akmeliawati, I. Abdulljabaar, Robust h∞ controller for high precision positioning system, design, analysis, and implementation, Intell. Control Autom. 03 (03) (2012) 262–273, http://dx.doi.org/10.4236/ica.2012.33030.

[51] J. Lunze, Regelungstechnik 1, Springer Berlin Heidelberg, Berlin, Heidelberg, 2014, http://dx.doi.org/10.1007/978-3-642-53909-1.

[52] H. Khalil, Nonlinear Control, Pearson, Boston, 2015.

[53] V. Utkin, H. Lee, Chattering problem in sliding mode control systems, IFAC Proc. Vol. 39 (5) (2006) 1, http://dx.doi.org/10.1016/b978-008044613-4.50002-1.

[54] R.E. Kalman, A new approach to linear filtering and prediction problems, J. Basic Eng. 82 (1) (1960) 35–45, http://dx.doi.org/10.1115/1.3662552.

[55] S.V. Drakunov, Sliding-mode observers based on equivalent control method, in: Proceedings of the 31st IEEE Conference on Decision and Control, IEEE, 1992, pp. 2368–2369, http://dx.doi.org/10.1109/cdc.1992.371368.

[56] SIEMENS AG, SINAMICS S120 function manual drive functions, 2004, URL: https://support.industry.siemens.com/cs/mdm/99686093?c=65848063883&lc=en-AE.

[57] K. Erkorkmaz, Y. Altintas, High speed CNC system design. Part I: jerk limited trajectory generation and quintic spline interpolation, Int. J. Mach. Tools Manuf. 41 (9) (2001) 1323–1345.

[58] K.L. Chien, On the automatic control of generalized passive systems, Trans. Asme 74 (1972) 175–185.

[59] T. Surmann, Simulation der Dynamik von Dreh-und Fräsprozessen, Vulkan Verlag, 2017.

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